WORKED SOLUTION

STEP I 2001, Question 12

Probability and Statistics • Tree Diagrams

STEP I 2001, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q12

The key idea

Everything depends on keeping careful track of *who knows what*. A student knows that someone has heard the rumour only if that person told them, or they told that person; nobody passes on where they got it from. Once we record each student's knowledge at every stage, the tree is small: every branching is a choice between two people, so it has only eight paths, and they all turn out to be equally likely.

Solution

Setting up

Write A, B, C, D for the four students. At any stage, the people whom student X *knows* have heard the rumour are: X themselves, whoever has told X the rumour, and whoever X has told it to. When X hears the rumour, X chooses uniformly at random among the people outside that set. If that set already contains all four students, the rumour stops with X.

At the start, A has told C, so A knows $\{A, C\}$ and C knows $\{A, C\}$.

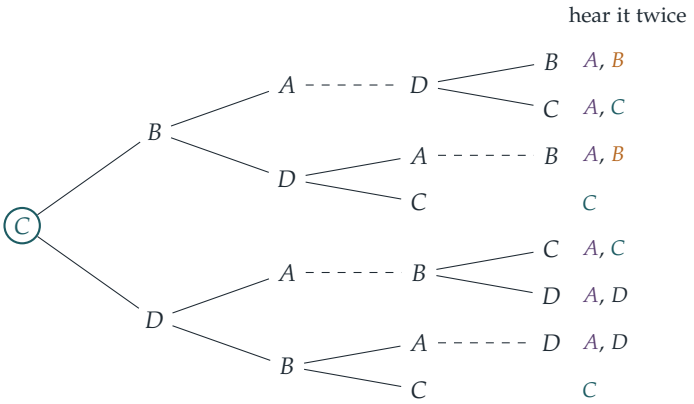
The tree

C omits A and tells B or D, each with probability $\frac{1}{2}$. Take the branch $C \to B$ first. Now C knows $\{A, B, C\}$ and B knows $\{B, C\}$ (not A, because C didn't say where the rumour came from). So B tells A or D, each with probability $\frac{1}{2}$.

- $B \to A$. Arthur now knows $\{A, B, C\}$, so he must tell D, and Arthur then knows everyone has heard. D knows only $\{A, D\}$, so D tells B or C with probability $\frac{1}{2}$ each. Either way the hearer completes their set: B already knew $\{A, B, C\}$, as did C, and each now learns of D. So the rumour stops.
- $B \to D$. Now B knows $\{B, C, D\}$ and D knows $\{B, D\}$, so D tells A or C.

- $D \to C$: C already knew $\{A, B, C\}$ and now learns of D, so the rumour stops. Arthur never rehears it.
- $D \to A$: Arthur knows $\{A, C, D\}$, so he tells B, who then knows all four. The rumour stops.

The branch $C \to D$ is the same with the roles of B and D swapped. Here is the whole tree. Every branch carries probability $\frac{1}{2}$ except the forced moves (probability 1), shown dashed.



Each path to a leaf contains exactly three choices with probability $\frac{1}{2}$ (the forced moves contribute a factor of 1), so each of the eight outcomes has probability $\frac{1}{8}$. The labels on the right show who hears the rumour twice on that path. (On every path a student hears it at most twice.)

Ask yourself Why does B, having just heard from C, not know that Arthur has heard it?

Because C doesn't say who told them. All B can deduce is that C has heard it. This is what keeps the rumour going: in the $C \to B \to A$ branch, Arthur is told something he already knew.

Arthur rehears it

Arthur rehears the rumour on every path except the two short ones, $C \to B \to D \to C$ and $C \to D \to B \to C$. So

$$\mathrm{P}(\text{Arthur rehears it}) = 1 - 2 \times \tfrac{1}{8} = \tfrac{3}{4},$$

as required.

Bertha and Chandra

Bertha hears it twice only on the paths $C \to B \to A \to D \to B$ and $C \to B \to D \to A \to B$, so

$$\mathrm{P}(\text{Bertha hears it twice}) = \tfrac{2}{8} = \tfrac{1}{4}.$$

Chandra hears it twice on four paths: two in each half of the tree ($C \to B \to A \to D \to C$, $C \to B \to D \to C$, $C \to D \to A \to B \to C$ and $C \to D \to B \to C$), so

$$\mathrm{P}(\text{Chandra hears it twice}) = \tfrac{4}{8} = \tfrac{1}{2}.$$

Chandra is more likely than Bertha to hear it again because Chandra heard it early: the people told later in the chain don't know that Chandra has heard it, so Chandra stays a possible target right to the end.

Checks

By symmetry between B and D, the probability that Delilah hears it twice should also be $\frac{1}{4}$, and the tree gives two such paths. The outcomes sum correctly: eight equally likely leaves, total probability 1. An exact enumeration of the process by computer gives $\frac{3}{4}$, $\frac{1}{4}$ and $\frac{1}{2}$, and a simulation of 200 000 rumours gave 0.750, 0.250 and 0.501.

Common slips

- Letting the hearer know everyone further back in the chain. Only the teller's identity is passed on, and with full knowledge the rumour would never return to Arthur.
- Forgetting that the *teller* also learns something: after telling Y, X knows that Y has heard it, which is what forces Arthur's moves.
- Stopping the tree as soon as Arthur rehears it. The later branches are needed for Bertha and Chandra.

Beyond STEP

The rumour is a random walk on the four students whose rules depend on the history, recorded here as each student's knowledge. Making the state rich enough that the next step depends only on the present state turns it into a *Markov chain*, the central object of a first course in random processes; rumour and epidemic models are a classic application. Grimmett and Welsh, *Probability: An Introduction*, covers Markov chains at exactly the right level.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.